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Installation standards: conduit, enclosures, heat and terminals

Solar ESS technical note · October 2026 · IMC, EMT, other conduit, combiner enclosures, heat dissipation, terminals

> **New in Thailand:** TISI approved มอก. 62852 (PV DC connectors) and มอก. 61439 เล่ม 8 (PV DC assemblies, meaning combiner boxes) as mandatory standards on 18 August 2026, expected in force from December 2026. Thai code rules below follow the EIT wiring code and the EIT rooftop PV standard, which match the NEC on most points.

Conduit

Type	Standard	Where it may be used	On-site check
IMC (intermediate metal)	มอก. 770-2533, UL 1242, ANSI C80.6	Indoor, outdoor, wet, exposed, embedded. Wall ¾" 2.10 mm, 1" 2.35 mm	มอก./UL print; caliper the wall; cold-galvanise cut threads; support every 3 m and within 0.9 m of boxes; no more than 360° of bends between pull points
EMT (thin wall)	มอก. 770, UL 797, ANSI C80.3	Indoor and dry locations. Not buried, not in slabs, not exposed on roofs	Compression or set-screw fittings rated for the location; ream cut ends
RSC (heavy wall)	มอก. 770, UL 6, ANSI C80.1	All locations including burial; best for exposed roof and coastal runs	Hot-dip galvanising
Flexible metal (FMC)	UL 1	Dry, short final drops only; never the earth path	Clip every 1.5 m and within 0.3 m of boxes
Liquid-tight flexible	UL 360	Wet locations; drops to inverter or battery	UV-rated listed glands; minimum bend radius
PVC / uPVC	Manufacturer / TIS	Standard PVC breaks down in UV; use UV-rated uPVC outdoors	Expansion fittings on hot roofs; supports every 0.9–1.5 m
HDPE	Manufacturer PN rating	Underground, buried at least 0.6 m	Bend radius ≥15× cable radius
Fill (all types)	EIT table, same as NEC	1 cable 53 %, 2 cables 31 %, 3 or more 40 %	Calculate from cable outer diameter

Combiner box enclosures

Item	Standard	Requirement	On-site check
Assembly	IEC 62208, IEC 61439-1/-2, IEC 61439-8	Temperature-rise verified; rating plate with DC voltage ≥ cold-weather Voc	Read the plate

Item	Standard	Requirement	On-site check
Ingress	IEC 60529	IP65 outdoors; IP66 at coastal or humid sites	Bottom entry only; glands tight; unused entries blanked
Impact	IEC 62262	IK08 or higher on roofs; IK10 in public areas	IK marking on datasheet
Material	—	UV-stabilised polycarbonate (Class II) or powder-coated steel (earthed); stainless or GRP at the coast	UV chalking, cracks, rust at hinges
DC equipment	IEC 62548-1:2023, IEC 60364-7-712	DC switch-disconnector to IEC 60947-3; string fuses 1.5–2.4 × Isc; SPD rated above Voc; PV warning labels	Load-break test; SPD status window; labels on box and isolator

Heat dissipation

Item	Requirement	On-site check
Hybrid inverter (Solis S6)	Single-phase: 250 mm clearance, 800 mm above floor, ambient ≤40 °C advised. Three-phase: 300 mm between units, 500 mm in front	Shade canopy where there is west sun; no attics or closed rooms; watch for temperature-derating events
LiFePO4 battery (Solis IntelliHome)	Operates −20 to 55 °C, best 10–45 °C, derates above 45 °C; not outdoors within 500 m of the coast	Check April ambient; keep out of the sun; nothing stacked on it
Breakers in boxes	IEC 60898 devices are calibrated at 30 °C; derate at 45–55 °C box temperature using the maker's table	Size the box to shed heat; shaded side; off hot roof sheet
Assemblies	IEC 61439: terminal temperature rise ≤70 K at 35 °C ambient	Thermal scan at full sun; a terminal 10–15 K hotter than similar ones needs attention

Terminals and connectors

Item	Standard	Requirement	On-site check
PV connectors	IEC 62852 / มอก. 62852	Mated pair must be the same type from the same maker	No mixed brands; maker's crimp tool; pull test
Crimp lugs	IEC 61238-1-1	Tinned copper; die matched to lug and tool; bimetallic lugs for aluminium	Die number on the crimp; lug size matches cable
Ferrules	Maker's data	On fine-stranded cable where the terminal requires; no solder-tinned ends	Checklist line for ferrules
Torque	Maker's values	e.g. Solis S6-EH3P: AC 4–5 N·m, PE 2 N·m	Calibrated driver; torque-mark every terminal
Temperature rating	Terminal 60/75/90 °C	Cable ampacity must not exceed the terminal rating	Read the marking on breaker or terminal

Most common mistakes we find in Thailand

EMT run exposed on roofs or cast into slabs · yellow PVC in the sun, cracking within 2–3 years · mixed MC4 brands and plier crimps, the main cause of DC arcing · IP65 boxes with open entries or top entry · AC breakers on the DC side, or fuses and SPDs rated below cold-weather Voc · inverters and batteries in direct sun or attics · no torque tool, no ferrules · conduit filled beyond 40 % · missing PV warning labels.

Specifications are taken from the manufacturers' published datasheets; check the current datasheet before ordering. For prices and a design for your site, book a consultation at www.solaress.org or LINE: solaress1.

Sources: IEC 62548-1:2023 · TISI PV standards, Aug 2026 · EIT rooftop PV standard summary · IMC submittal · Connector cross-mating

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